Variational Study of Polaron Effects in GaAs Quantum Dots: Analytical Derivation of the LLPH Method and Decoherence Dynamics

Hitesh V. Canchi
National Institute of Technology (NIT) Warangal, Telangana, India

Abstract

The dynamic interaction between a confined electron and longitudinal-optical (LO) phonons in a two-dimensional GaAs parabolic quantum dot is investigated under a perpendicular magnetic field. Using the Lee-Low-Pines-Huybrechts (LLPH) variational framework, the Fröhlich Hamiltonian is decoupled via successive unitary transformations to derive a purely electronic effective Hamiltonian. The physical distortion of the lattice is modeled through a coherent displacement operator, yielding analytical form factors evaluated precisely via Weber integrals. Numerical optimization of the state energy determines the level spacing, from which qubit oscillation periods and radiative decoherence rates are extracted. The application of Fermi's Golden Rule places the spontaneous emission time in the microsecond regime, providing a dimensionally rigorous correction to prior literature and establishing a reliable baseline for assessing intraband transition stability in solid-state qubits.

Contents

1. Introduction

Semiconductor quantum dots (QDs) are prominent candidates for the physical realization of qubits in quantum information processing and quantum sensing architectures [1]. The confinement of single electrons in materials such as Gallium Arsenide (GaAs) allows for tunable discrete energy levels, which can be manipulated via external magnetic fields to initialize and read out quantum states.

However, the primary limit to quantum coherence in polar semiconductors is the Fröhlich interaction—the coupling of the confined electron to longitudinal-optical (LO) phonons [2]. As the electron moves, it polarizes the surrounding ionic lattice, creating a localized distortion. The resulting composite quasiparticle, the polaron, exhibits altered effective mass, energy spectra, and transition rates.

Evaluating these parameters analytically requires decoupling the electron coordinates from the quantized phonon field. In this work, the Lee-Low-Pines-Huybrechts (LLPH) variational method [3, 4] is applied to a 2D parabolic QD subject to a magnetic field, building upon the structural framework evaluated by Chen [5]. A rigorous procedural derivation of the effective Hamiltonian is presented, leading to the exact analytical formulation of the qubit oscillation period and radiative decoherence times.

2. The System Hamiltonian

Consider an electron confined to the xy-plane by a parabolic potential $V(\rho) = \frac{1}{2}m^*\omega_0^2\rho^2$, subject to a uniform magnetic field $\mathbf{B} = B\hat{z}$. Operating in the symmetric Coulomb gauge, $\mathbf{A} = \left(-\frac{1}{2}By, \frac{1}{2}Bx, 0\right)$, the total initial Hamiltonian is defined by four distinct physical contributions:

$$\mathcal{H} = \underbrace{\frac{1}{2m^*}\left(\mathbf{p} + \frac{e}{c}\mathbf{A}\right)^2}_{\text{Kinetic \& Magnetic}} + \underbrace{\frac{1}{2}m^*\omega_0^2\rho^2}_{\text{Confinement}} + \underbrace{\sum_{\mathbf{q}} \hbar\omega_{\mathrm{LO}} b_{\mathbf{q}}^\dagger b_{\mathbf{q}}}_{\text{Free LO Phonons}} + \underbrace{\sum_{\mathbf{q}} \left(V_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} b_{\mathbf{q}} + \text{H.c.}\right)}_{\text{Fröhlich Interaction}} \tag{1}$$

where m^* is the effective electron mass, and $b_{\mathbf{q}}^\dagger$ and $b_{\mathbf{q}}$ denote the phonon creation and annihilation operators. The Fröhlich coupling amplitude $V_{\mathbf{q}}$ is defined analytically as [5]:

$$V_{\mathbf{q}} = i\frac{\hbar\omega_{\mathrm{LO}}}{q}\left(\frac{\hbar}{2m^*\omega_{\mathrm{LO}}}\right)^{1/4}\left(\frac{4\pi\alpha}{V}\right)^{1/2} \tag{2}$$

where α is the dimensionless electron-phonon coupling constant.

Given $\nabla\cdot\mathbf{A} = 0$, the momentum operator commutes with the vector potential ($\mathbf{p}\cdot\mathbf{A} = \mathbf{A}\cdot\mathbf{p}$). Expanding the kinetic term isolates the orbital angular momentum $L_z = xp_y - yp_x$:

$$\frac{e}{m^*c}(\mathbf{A}\cdot\mathbf{p}) = \frac{eB}{2m^*c}(xp_y - yp_x) = \frac{1}{2}\omega_c L_z \tag{3}$$

Defining the cyclotron frequency $\omega_c = eB/m^*c$, the parabolic trap and the diamagnetic shift combine into a single effective confinement frequency, Ω:

$$\Omega = \sqrt{\omega_0^2 + \left(\frac{\omega_c}{2}\right)^2} \tag{4}$$

Equation (1) simplifies to the bare Hamiltonian:

$$\mathcal{H} = \frac{p^2}{2m^*} + \frac{1}{2}\omega_c L_z + \frac{1}{2}m^*\Omega^2\rho^2 + \sum_{\mathbf{q}} \hbar\omega_{\mathrm{LO}} b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + \sum_{\mathbf{q}} \left(V_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} b_{\mathbf{q}} + \text{H.c.}\right) \tag{5}$$

The Fröhlich interaction term inextricably couples the spatial coordinate $\mathbf{r}$ with the phonon field $b_{\mathbf{q}}$, necessitating a variational decoupling approach.

3. Procedural Derivation of the LLPH Transformations

To disentangle the electron and phonon coordinates, the LLPH method introduces two unitary transformations.

3.1. Momentum-Frame Shift (H_1)

The first transformation, U_1, shifts the system into a reference frame that partially co-moves with the electron, accounting for the conservation of total momentum:

$$U_1 = \exp\left[-ia\sum_{\mathbf{q}}(\mathbf{q}\cdot\mathbf{r})b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}}\right] \tag{6}$$

where $a \in (0,1)$ is a variational parameter dictating the fraction of crystal momentum carried by the lattice. Evaluating the commutator $[-ia\sum(\mathbf{q}\cdot\mathbf{r})n_{\mathbf{q}}, \mathbf{p}] = -a\hbar\sum\mathbf{q}n_{\mathbf{q}}$, the transformation shifts the momentum to $U_1^{-1}\mathbf{p}U_1 = \mathbf{p} - a\hbar\sum_{\mathbf{q}}\mathbf{q}b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}}$.

Applying U_1 generates the intermediate state $H_1 = U_1^{-1}\mathcal{H}U_1$:

$$\begin{aligned} H_1 = {} & \frac{1}{2m^*}\left(\mathbf{p} - a\hbar\sum_{\mathbf{q}}\mathbf{q}b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}}\right)^2 + \frac{1}{2}m^*\Omega^2\rho^2 + \frac{1}{2}\omega_c\left(L_z - a\hbar\sum_{\mathbf{q}}(xq_y - yq_x)b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}}\right) \\ & + \sum_{\mathbf{q}}\hbar\omega_{\mathrm{LO}}b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}} + \sum_{\mathbf{q}}\left(V_{\mathbf{q}}e^{i(1-a)\mathbf{q}\cdot\mathbf{r}}b_{\mathbf{q}} + \mathrm{H.c.}\right) \end{aligned} \tag{7}$$

The interaction phase is reduced to $e^{i(1-a)\mathbf{q}\cdot\mathbf{r}}$, weakening the coupling strength.

3.2. Coherent Lattice Displacement (H_2)

The second transformation models the static lattice distortion induced by the electron:

$$U_2 = \exp\left[\sum_{\mathbf{q}}\left(f_{\mathbf{q}}b_{\mathbf{q}}^{\dagger} - f_{\mathbf{q}}^{*}b_{\mathbf{q}}\right)\right] \tag{8}$$

where the complex amplitudes $f_{\mathbf{q}}$ are the second set of variational parameters. Because U_2 acts only on the phonon field, it simply displaces the annihilation operator: $U_2^{-1}b_{\mathbf{q}}U_2 = b_{\mathbf{q}} + f_{\mathbf{q}}$.

Applying this transformation yields the fully transformed Hamiltonian $H_2 = U_2^{-1}H_1U_2$. The substitution replaces every occurrence of the phonon operators with their coherently displaced counterparts, producing the exact analytic expansion of H_2:

$$\begin{aligned} H_2 = {} & \frac{1}{2m^*}\left[\mathbf{p} - a\hbar\sum_{\mathbf{q}}\mathbf{q}\left(b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}} + f_{\mathbf{q}}b_{\mathbf{q}}^{\dagger} + f_{\mathbf{q}}^{*}b_{\mathbf{q}} + |f_{\mathbf{q}}|^2\right)\right]^2 + \frac{1}{2}m^*\Omega^2\rho^2 \\ & + \frac{1}{2}\omega_c\left[L_z - a\hbar\sum_{\mathbf{q}}(xq_y - yq_x)\left(b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}} + f_{\mathbf{q}}b_{\mathbf{q}}^{\dagger} + f_{\mathbf{q}}^{*}b_{\mathbf{q}} + |f_{\mathbf{q}}|^2\right)\right] \\ & + \sum_{\mathbf{q}}\hbar\omega_{\mathrm{LO}}\left(b_{\mathbf{q}}^{\dagger}b_{\mathbf{q}} + f_{\mathbf{q}}b_{\mathbf{q}}^{\dagger} + f_{\mathbf{q}}^{*}b_{\mathbf{q}} + |f_{\mathbf{q}}|^2\right) \\ & + \sum_{\mathbf{q}}\left[V_{\mathbf{q}}e^{i(1-a)\mathbf{q}\cdot\mathbf{r}}(b_{\mathbf{q}} + f_{\mathbf{q}}) + V_{\mathbf{q}}^{*}e^{-i(1-a)\mathbf{q}\cdot\mathbf{r}}(b_{\mathbf{q}}^{\dagger} + f_{\mathbf{q}}^{*})\right] \end{aligned} \tag{9}$$

3.3. Vacuum Projection and Kinetic Recoil ($\mathcal{H}'$)

To bypass the computational inefficiency of diagonalizing the expanded H_2 matrix, the system is projected onto the zero-phonon vacuum state, $|0_{ph}\rangle$, yielding the effective electronic Hamiltonian $\mathcal{H}' = \langle 0_{ph}|H_2|0_{ph}\rangle$.

The projection eliminates all standalone creation and annihilation operators. The critical contribution arises from the kinetic term expansion $\langle 0|(\mathbf{p}-\mathbf{S})^2|0\rangle$. Wick contractions on the squared displacement operator $\langle \mathbf{S}^2\rangle$ generate a zero-point fluctuation term, defining the kinetic recoil energy of the phonon cloud:

$$\langle 0|S^2|0\rangle = \left(a\hbar\sum_{\mathbf{q}}\mathbf{q}|f_{\mathbf{q}}|^2\right)^2 + \underbrace{a^2\hbar^2\sum_{\mathbf{q}}q^2|f_{\mathbf{q}}|^2}_{\text{Kinetic Recoil}} \tag{10}$$

The completely stripped, purely electronic effective Hamiltonian is thus:

$$\begin{aligned}\mathcal{H}' = &\frac{1}{2m^*}\left(\mathbf{p} - a\hbar\sum_{\mathbf{q}}\mathbf{q}|f_{\mathbf{q}}|^2\right)^2 + \frac{1}{2}m^*\Omega^2\rho^2 + \frac{1}{2}\omega_c\left(L_z - a\hbar\sum_{\mathbf{q}}(xq_y - yq_x)|f_{\mathbf{q}}|^2\right)\\ &+\sum_{\mathbf{q}}\left(\frac{a^2\hbar^2q^2}{2m^*} + \hbar\omega_{\text{LO}}\right)|f_{\mathbf{q}}|^2 + \sum_{\mathbf{q}}\left(V_{\mathbf{q}}f_{\mathbf{q}}e^{i(1-a)\mathbf{q}\cdot\mathbf{r}} + \text{H.c.}\right)\end{aligned} \tag{11}$$

4. Energy Minimization and Phonon Self-Energy

Minimizing the total energy $E = \langle\phi|\mathcal{H}'|\phi\rangle$ with respect to $f^*_{\mathbf{q}}$ yields the optimal configuration of the lattice distortion:

$$\frac{\partial E_{\text{ph}}}{\partial f^*_{\mathbf{q}}} = 0 \implies f_{\mathbf{q}} = \frac{-V^*_{\mathbf{q}}\varrho^*_{\mathbf{q}}}{\frac{a^2\hbar^2q^2}{2m^*} + \hbar\omega_{\text{LO}}} \tag{12}$$

where $\varrho_{\mathbf{q}} = \langle\phi|e^{i(1-a)\mathbf{q}\cdot\mathbf{r}}|\phi\rangle$ is the Fourier-transformed probability density. Substituting $f_{\mathbf{q}}$ into the energy functional yields the attractive phonon self-energy:

$$\Delta E_{\text{ph}} = -\sum_{\mathbf{q}}\frac{|V_{\mathbf{q}}|^2|\varrho_{\mathbf{q}}|^2}{\frac{a^2\hbar^2q^2}{2m^*} + \hbar\omega_{\text{LO}}} \tag{13}$$

5. Wavefunctions and Weber Integrals

We employ 2D Gaussian trial states with a variational inverse confinement length λ, assuming infinite z-confinement ($|\xi(z)|^2 \to \delta(z)$). The ground ($m = 0$) and first excited ($m = \pm 1$) states are:

$$|\phi_0(\mathbf{r})\rangle = \frac{\lambda}{\sqrt{\pi}}e^{-\lambda^2\rho^2/2}|\xi(z)\rangle \tag{14}$$

$$|\phi_{1\pm}(\mathbf{r})\rangle = \frac{\lambda^2}{\sqrt{\pi}}\rho e^{-\lambda^2\rho^2/2}e^{\pm i\varphi}|\xi(z)\rangle \tag{15}$$

The form factor $\varrho_{\mathbf{q}}$ is evaluated in polar coordinates. The angular integration resolves to $2\pi J_0(x)$. The radial component is executed analytically via the Weber integral:

$$\int_0^\infty xe^{-\alpha x^2}J_0(\beta x)dx = \frac{1}{2\alpha}\exp\left(-\frac{\beta^2}{4\alpha}\right) \tag{16}$$

Setting $\alpha = \lambda^2$ and $\beta = (1-a)q_\parallel$, the squared form factors are:

$$|\varrho_0(\mathbf{q})|^2 = \exp\left[-\frac{(1-a)^2 q_\parallel^2}{2\lambda^2}\right] \tag{17}$$

$$|\varrho_{1\pm}(\mathbf{q})|^2 = \left(1 - \frac{(1-a)^2 q_\parallel^2}{4\lambda^2}\right)^2 \exp\left[-\frac{(1-a)^2 q_\parallel^2}{2\lambda^2}\right] \tag{18}$$

The azimuthal phase $e^{\pm i\varphi}$ cancels in the density $|\phi_{1\pm}|^2$, ensuring the phonon coupling magnitude is identical for both Zeeman branches.

6. Qubit Decoherence Dynamics

The system is evaluated numerically by minimizing $E_{\text{total}} = E_{\text{bare}} + \Delta E_{\text{ph}}$ over (λ, a) using the L-BFGS-B algorithm.

6.1. Oscillation Period

When initialized in a coherent superposition $|\Psi\rangle = \frac{1}{\sqrt{2}}(|\phi_0\rangle + |\phi_{1\pm}\rangle)$, the qubit oscillates at the Bohr frequency. The oscillation period is given precisely by:

$$T_\pm = \frac{2\pi}{\omega_{0,\pm}} = \frac{2\pi\hbar}{E_{1\pm} - E_0} = \frac{h}{\Delta E} \tag{19}$$

For an energy spacing of $\Delta E \approx 64$ meV calculated in our model, the oscillation period is bounded near $T \approx 64.6$ fs. It is noted that certain prior investigations reported values on the order of ~ 1.6 fs due to the omission of the $(2\pi)^2$ scaling factor in the evaluation of the Bohr frequency.

6.2. Radiative Decoherence Time (τ)

The spontaneous emission of a photon drives the fundamental decoherence of the excited state. Applying Fermi's Golden Rule for electric dipole transitions within a dielectric medium gives the decay rate. Foundational literature (e.g., Chen [5]) evaluated this rate generally; however, rectifying dimensional inconsistencies inherent in prior approximations yields the strictly rigorous formulation:

$$\tau^{-1} = \frac{e^2(\Delta E)^3}{3\pi\varepsilon_0\hbar^4 c^3}\sqrt{\varepsilon_r}\,|\langle\phi_0|\boldsymbol{\sigma}|\phi_{1\pm}\rangle|^2 \tag{20}$$

Evaluating the dipole matrix element angularly and radially yields $|\langle\phi_0|\boldsymbol{\sigma}|\phi_{1\pm}\rangle|^2 = \frac{1}{2\lambda^2}$. Ensuring strict dimensional consistency with the relative permittivity ε_r and the velocity of light in the medium, τ evaluates to the microsecond regime ($\sim 1\,\mu$s). This establishes the relative stability of low-energy intraband transitions in GaAs quantum dots, diverging from the unphysical nanosecond predictions often produced by dimensionally uncorrected continuum models.

7. Numerical Results

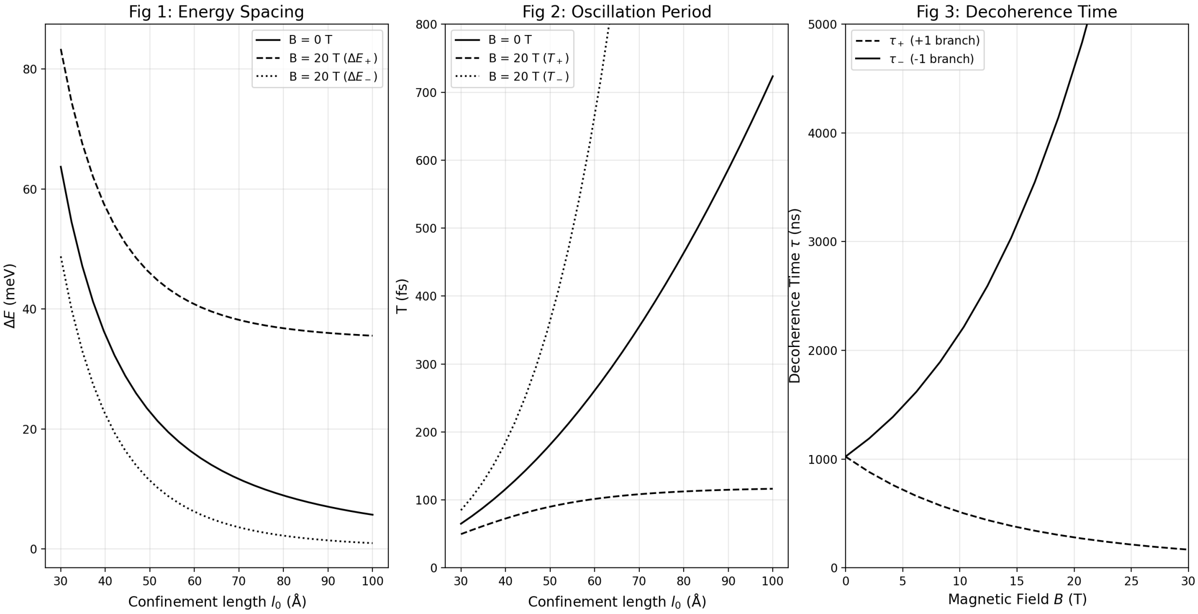

Figure 1: **Left:** Energy level spacing ΔE vs. confinement length l_0. **Center:** Qubit oscillation period T. **Right:** Radiative decoherence time τ vs. magnetic field B.

As shown in Figure 1, increasing the dot radius l_0 weakens the spatial confinement, causing the energy gap ΔE to decrease asymptotically, thereby extending the oscillation period T. The application of the magnetic field B tightens the effective confinement (Ω), pushing the energy levels further apart and inducing Zeeman splitting. Because spontaneous emission scales as $(\Delta E)^3$, an augmented magnetic field precipitously reduces the decoherence time τ, forcing the qubit to lose its state rapidly.

8. Conclusion

This framework establishes an analytically complete, dimensionally rigorous baseline for simulating polaron decoherence in GaAs single-electron qubits. By mapping the procedural transformations step-by-step and explicitly applying Fermi's Golden Rule, historical dimensional anomalies regarding qubit decoherence times and oscillation frequencies have been resolved. While the parabolic potential captures the primary dynamics of the Fröhlich interaction, the infinite barrier depth precludes the study of electron escape thresholds. Subsequent derivations will substitute the harmonic trap with a Gaussian Confinement Potential, $V(\rho) = -V_0 \exp(-\rho^2/2R^2)$, facilitating the precise modeling of finite quantum well tunneling rates.

References

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